



Band-Split Self-supervised Mamba for Infant-centered Audio Analysis

Xulin Fan^{1,2}, Jialu Li², Mark Hasegawa-Johnson^{1,2}, Nancy L. McElwain^{2,3}

¹Department of Electrical and Computer Engineering, University of Illinois, USA

²Beckman Institute for Advanced Science and Technology, University of Illinois, USA

³Department of Human Development and Family Studies, University of Illinois, USA

xulinf2@illinois.edu, jialuli3@illinois.edu, jhasegaw@illinois.edu, mcelwn@illinois.edu

Abstract

Infant-worn audio recorders provide a valuable means to analyze an infant's home environment and vocal interactions with family members. Recent advances in self-supervised learning on large unlabeled datasets and supervised training on limited annotated data have improved performance in this domain. However, data scarcity remains a challenge. We introduce Band-Split SSAMBA (BS-SSAMBA), a self-supervised representation learning method that incorporates band-specific projections and a band-agnostic Mamba encoder to model temporal relationships across frequency bands. Designed for data-efficient learning, BS-SSAMBA effectively leverages both unlabeled and labeled in-domain data. Through extensive experiments on family audio recordings, we show that BS-SSAMBA outperforms vanilla SSAMBA and wav2vec2-based models, demonstrating its effectiveness for infant-centered audio tagging.

Index Terms: infant-centered audio analysis, speaker diarization, vocalization classification, self-supervised learning

1. Introduction

Infants communicate a wide range of needs and emotions through vocalizations, such as cries, fusses, and babbles. Caregivers' timely and appropriate responses to infants' vocal cues [1] and infant-caregiver vocal turn-taking [2] [3] have important implications for social-emotional [4] [5], cognitive [6], and language [7] development. Because infants and mothers show meaningful differences in behavior, including vocalizations, across lab and home contexts [8] [9] and because father-infant [10] and sibling-infant interactions [11] also play important roles in infant development, automated diarization of infants' and family members' vocalizations across daylong home recordings has the potential to transform scientific understanding of these early yet critical developmental processes. To accomplish such diarization tasks, audio tagging has emerged as a powerful tool for labeling audio segments or frames with information about who is speaking (e.g., infant, adult) and the type of vocalization (e.g., crying, babbling). Previous works on infant-centered audio have largely relied on fully supervised approaches, such as infant vocalization classification [12, 13, 14], cry classification [15], and speaker diarization [16], but these methods often suffer from limited performance due to the scarcity of labeled data.

Recent advances in self-supervised learning (SSL) are transforming how we tackle audio tagging, particularly in domains with limited labeled data. By leveraging large-scale unlabeled audio, models such as wav2vec2.0 [17], the Audio Spectrogram Transformer (AST) [18], and Self-Supervised Audio Mamba (SSAMBA) [19] learn high-level representations before

fine-tuning on small labeled datasets. Compared with purely supervised approaches, these SSL methods substantially reduce the amount of labeled data needed and often yield better performance. Furthermore, spectrogram-based methods like AST (and its self-supervised variant SSAST [20]) capture frequency-time structures effectively, yet often require large-scale data and can be computationally expensive when modeling long sequences. On the other hand, SSAMBA uses a state space model [21] that scales linearly with sequence length, addressing the complexity and memory challenges associated with long audio contexts.

Despite these advances, audio tagging for infant-centered home audio recordings remains under-explored due to the scarcity of in-domain data. Some recent works using infant-centered recordings have begun to bridge this gap. For instance, Khan et al. [22] employed a pretrained AST model for environmental sound detection in the home, whereas Li et al. [23] leveraged wav2vec2.0 pretraining and finetuning schemes to solve speaker diarization and vocalization classification for infants and parents. Both studies demonstrated the potential benefits of SSL in infant-centered tasks but faced challenges like model complexity and limited domain-specific data.

In this paper, we propose a new method, Band-Split SSAMBA (BS-SSAMBA), to further advance SSL of audio representation for infant-centered family audio tagging. Below we outline our major contributions:

1. We introduce Band-Split SSAMBA, a state space model (SSM) that disentangles spectrogram modeling into frequency modeling via band-specific projections and temporal modeling via a Mamba encoder. This design reduces the sequence length and difficulty of spectrogram modeling for the mamba encoder and facilitates better representation learning with limited in-domain data.

2. We incorporate framewise feature conditioning, making important cross-band information easily accessible to the model during fine-tuning, improving efficiency while minimizing computational overhead.

3. We conduct extensive experiments on unlabeled and labeled infant-centered audio recordings in the home. Through comparison with existing SSL methods, we validate the effectiveness of our band-split mechanism and framewise feature conditioning.

2. Method

Figure 1 shows our model architecture. Our approach builds upon SSAMBA [19], a self-supervised SSM designed for general-purpose audio tagging. SSAMBA retains the pretraining and fine-tuning objectives of the earlier transformer-based SSAST model [20] but achieves greater efficiency due to the

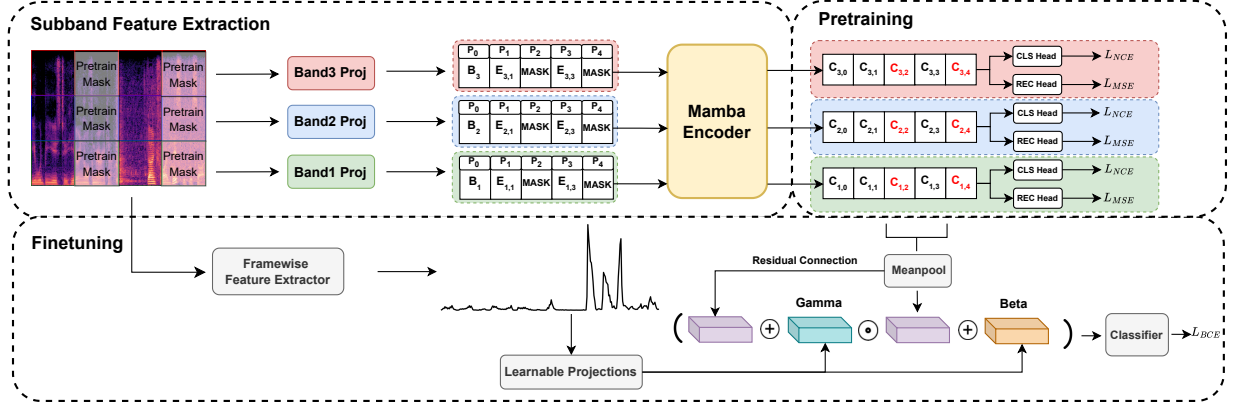


Figure 1: Pretraining and Finetuning stages of the Band-split SSAMBA Architecture

linear-time complexity of SSMs. However, SSAMBA still struggles with data scarcity, particularly when both pretraining and fine-tuning datasets are significantly smaller than those used in general-purpose audio tagging.

To mitigate the effects of limited data and improve performance, we propose a hybrid approach that combines band-specific convolutional kernels and a unified temporal Mamba encoder, which is shared across frequency bands. This design enhances data efficiency and strengthens feature learning, making the model more robust in the case of limited in-domain data. Furthermore, we propose to incorporate framewise feature conditioning to mitigate the model’s lack of cross-band modeling ability.

2.1. Spectrogram Representation

The input audio waveform is first transformed into a 128-dimensional mel-spectrogram, providing a time-frequency representation of the signal. This transformation is performed using the Short-Time Fourier Transform (STFT) with a 25 ms window and a 10 ms stride, producing a spectrogram matrix X of dimensions $F \times T$, where F represents the number of frequency bins and T denotes the number of time frames. The spectrogram is then partitioned into overlapping subbands, resulting in a final structure of dimensions $B \times K \times T$, where B is the number of subbands and K is the number of frequency bins per subband. Finally, each subband spectrogram is further divided into 2D patches, forming patch sequences for downstream processing.

2.2. Band-specific Projection

To ensure that the Mamba encoder processes patch sequences from different subbands consistently, we employ linear projection layers with independent weights for each band. These layers map patches from different subbands into a unified embedding space, enabling a coherent representation across all subbands.

2.3. Mamba Encoder

To prepare the patch sequences for input into the Mamba encoder, we prepend band-specific tokens to each sequence to indicate the corresponding band and apply a shared positional embedding. Specifically, for a patch sequence from band b , denoted as $[E_{b,1}, E_{b,2}, E_{b,3}, \dots, E_{b,T}]$, we prepend a band-specific token B_b and add a learnable positional embedding

$[P_0, P_1, P_2, \dots, P_T]$ to the sequence. The final input sequence to the Mamba encoder is then represented as:

$$[B_b + P_0, E_{b,1} + P_1, E_{b,2} + P_2, \dots, E_{b,T} + P_T]$$

The band token B_b allows Mamba to learn band-specific information, while the positional embedding P_t helps Mamba to learn temporal relationships. Since temporal structures are consistent across different bands, we use a shared positional embedding sequence for all bands.

The Mamba encoder consists of stacked bidirectional SSMs as proposed in Vision Mamba [24]. The bidirectional SSM processes patch sequences in both forward and backward directions, effectively capturing temporal dependencies from both perspectives.

2.4. Training Objectives

During pretraining, we apply both generative and discriminative losses to patch sequences from each of the B frequency bands. We randomly select N indices from the total S patch positions in each band, denoted by the set $\mathcal{M} \subset \{1, \dots, S\}$, where S is the number of patches per band and $N = |\mathcal{M}|$. The patches at these selected positions are masked across all bands. The masked tokens’ contextualized embeddings from the mamba encoder are passed to two prediction heads: a generative head and a classification head. The output of these heads are optimized via Mean Squared Error (MSE) (L_g) and InfoNCE losses (L_d), respectively.

We define the combined loss as:

$$L = L_d + \lambda L_g,$$

where

$$L_d = -\frac{1}{BN} \sum_{b=1}^B \sum_{s \in \mathcal{M}} \log \left(\frac{\exp(\hat{\mathbf{c}}_{b,s} \cdot \mathbf{e}_{b,s})}{\sum_{j \in \mathcal{M}} \exp(\hat{\mathbf{c}}_{b,j} \cdot \mathbf{e}_{b,j})} \right),$$

$$L_g = \frac{1}{BN} \sum_{b=1}^B \sum_{s \in \mathcal{M}} \|\hat{\mathbf{x}}_{b,s} - \mathbf{x}_{b,s}\|^2.$$

Here, $\mathbf{x}$, $\mathbf{e}$, $\hat{\mathbf{c}}$, and $\hat{\mathbf{x}}$ respectively denote the patch, the projected patch embedding, the output of the classification head, and the output of the generative head. The subscripts b, s indicate the s -th patch along the temporal axis at band b .

During fine-tuning, we form a single embedding $\bar{\mathbf{c}}_i$ for the i th sample by meanpooling $\hat{\mathbf{c}}_{b,s}$ across all bands in all the column that is part of the labeled region. After incorporating the

SPLIT	# OF FAMILIES	TOTAL	SIL	CXN	F-ADS	F-CDS	M-ADS	M-CDS	FM-SNG	C-BAB	C-CRY	C-FUS
TRAINING	51	610.0	378.4	27.1	37.6	48.9	18.4	13.5	7.9	50.2	6.8	21.2
VALIDATION	5	105.0	72.5	1.8	5.5	5.5	3.1	1.2	1.0	6.6	0.2	7.6
TESTING	5	132.9	89.8	1.6	5.3	7.0	4.5	3.2	1.6	7.3	4.9	7.6

Table 1: Detailed statistics of training, validation, and testing split for annotated data. Each column denotes the total durations for each class in minutes. Columns from left to right are number of families, total, silence (SIL), siblings (CXN), adult/child-directed speech of female adult (F-ADS/F-CDS), adult/child-directed speech of male adults (M-ADS/M-CDS), Singing/Rhythmic Speech of adults (FM-SNG), babbling (C-BAB), crying (C-CRY), and fussing (C-FUS) of focal child.

framework feature conditioning (discussed in Section 2.5), this embedding is fed into the final classifier.

2.5. Framework Feature Conditioning

Although we successfully reduce the sequence length through band splitting, thereby decreasing the complexity of modeling long sequences, this approach also limits the model’s ability to capture cross-band relationships. For the audio tagging task of interest, certain framework characteristics play a crucial role in distinguishing different classes in the home audio recordings. For example, because the recording device is worn by the infant, the distance to adult caregivers—and consequently, the volume of their voices—serves as a strong cue for distinguishing whether an adult is speaking to another adult or to the infant.

Because Band-Split SSAMBA has a straightforward mechanism for cross-band modeling, it may struggle to efficiently extract framework features across all bands. To address this, we explicitly extract the framework energy of the spectrogram and use it as an additional conditioning cue for the final classifier during fine-tuning. We adopt Featurewise Linear Modulation (FiLM, [25]) as our conditioning mechanism, which can be formulated as:

$$\gamma_i = f(\mathcal{E}_i) \quad \beta_i = g(\mathcal{E}_i)$$

$$F(\bar{\mathbf{c}}_i | \mathcal{E}_i) = (1 + \gamma_i) \odot \bar{\mathbf{c}}_i + \beta_i.$$

where $\bar{\mathbf{c}}_i$ and $\mathcal{E}_i$ are 1-D vectors corresponding to the mean-pooled output of the Mamba encoder across patches in the central annotation area across all bands and the precomputed framework energy of the i -th sample, respectively, and $\odot$ denotes array multiplication. The functions f and g are learnable mappings that transform the framework energy into modulation parameters, which are then applied to the Mamba encoder’s output after meanpooling denoted as $\bar{\mathbf{c}}_i$. These modulation parameters, γ_i and β_i , act as an adaptive affine transformation F on $\bar{\mathbf{c}}_i$. The transformed embedding is subsequently fed into the final classification layer, with a residual connection from the original embedding retained.

3. Experiments

In this section, we present our experiments and dataset. We first describe our sample and data collection procedures, followed by our experimental results, comparisons with baseline methods, and ablations studies.

3.1. Data

For the experiments, we use audio recordings from 61 families with infants between 1 and 18 months of age (Mean=9.43 months). Families were recruited from a midsized midwestern city in the United States and were predominantly White and native English speakers. Multiple daylong home audio recordings (2-3 per family) were collected using LittleBeats™ [26], an infant wearable device that records audio, cardiac, and motion

METHOD	PATCH	STRIDE	ACC	MACRO-F1	KAPPA
W2V-LB [23]	NA	NA	0.814	0.599	0.655
SSAMBA [19]	16x16	10x10	0.792	0.529	0.599
SSAMBA [†]	16x16	16x16	0.798	0.533	0.620
SSAMBA [†]	64x4	64x4	0.819	0.581	0.647
SSAMBA [†]	64x4	32x4	0.804	0.554	0.633
BS-SSAMBA	64x4	32x4	0.825	0.609	0.663

Table 2: Main results and comparison between our proposed methods and baselines. For SSAMBA, STRIDE denotes the stride during finetuning only. For BS-SSAMBA, STRIDE denotes both pretraining and finetuning stride as a result of band split. [†] denotes the SSAMBA variants which have different patch and stride setting than the one used in [19].

data on a single platform.

Following prior studies [23], our model undergoes pretraining on a large unlabeled dataset before fine-tuning on a small set of labeled data. For pretraining, we use the same unlabeled dataset as in [23], comprising approximately 1,100 hours of LittleBeats recordings. These recordings are segmented into non-overlapping 10-second samples.

For supervised fine-tuning, we use around 14 hours of labeled data from our sample of 61 families. Audio files varying from 30 to 120 seconds in length were annotated for infant, adult, and sibling vocalizations by trained, certified research assistants using Praat software. The labeled dataset is divided into training, validation, and test sets, with the distribution of each split shown in Table 1. To ensure generalization, the families in each split are mutually exclusive. Each labeled sample is a 10-second audio clip, where annotations specify the active speaker and vocalization type within the central 0.1-second interval. Such long context was proved beneficial for accurate frame-level tagging for naturalistic home recordings during our preliminary experiments. Tags are drawn from a set of 10 labels: {SIL, CXN, F-ADS, F-CDS, M-ADS, M-CDS, FM-SNG, C-BAB, C-CRY, C-FUS} with class names described in the caption of Table 1.

3.2. Experimental Configuration

We pretrain the models for 15 epochs using the Adam optimizer with a learning rate of 1×10^{-4} and a batch size of 32. We set $\lambda = 10$ as the weighting term for the pretraining loss and $N = 192$ as the number of masked timesteps. Training was done on 4 NVIDIA A100 GPUs where pretraining takes around 3 days. After pretraining, we fine-tune the model using binary cross-entropy loss with mixup training [27] (mixup ratio = 0.6) and SpecAugment (maximum frequency mask length = 48 frames). The same optimizer is used for fine-tuning. The model is fine-tuned for five epochs, and we select the checkpoint with the best kappa score for evaluation. We use the base configuration from SSAMBA [19] for our mamba encoder, consisting of 24 layers with an embedding dimension of 768.

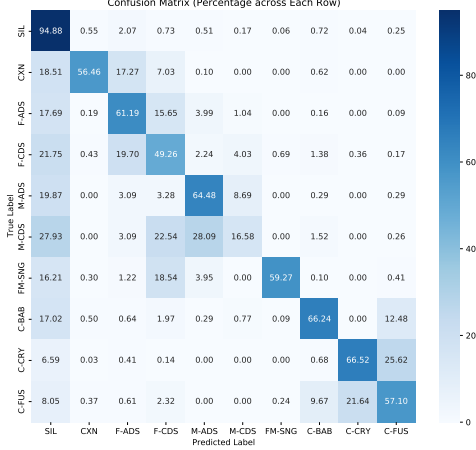


Figure 2: Confusion Matrix for BS-SSAMBA for Test set

METHOD	FRAME COND	ACC	MACRO-F1	KAPPA
BS-SSAMBA	No	0.821	0.597	0.656
BS-SSAMBA	Yes	0.825	0.609	0.663

Table 3: Ablation study on framewise feature conditioning. FRAME COND denotes whether we include framewise feature conditioning during the finetuning stage.

3.3. Main Results

We present our main results and comparisons with baseline methods in Table 2. As baselines, we include Wav2Vec2-LittleBeats (W2V-LB) [23] and SSAMBA [19]. All the models we compare have similar size: W2V-LB has 95M parameters, SSAMBA 101M parameters, BS-SSAMBA 102M parameters.

For W2V-LB, we use the official released checkpoint, which was pretrained on the same unlabeled dataset, and then fine-tuned using our labeled data. The SSAMBA models are pretrained and finetuned with most of the hyperparameters as released in [19], except number of masked patches and data augmentation techniques. We evaluate variants with different patch sizes while keeping the percentage of masked patches fixed at 75% and use the same specaugment and mixup training technique as used by BS-SSAMBA. We find that the default 16x16 patch size yields suboptimal performance for both SSAMBA and BS-SSAMBA. The primary reason for this performance drop appears to be the lack of temporal resolution, which is crucial for our downstream task requiring label resolution of 10 frames (0.1s).

From the main results, we observe that BS-SSAMBA outperforms both W2V-LB and all tested SSAMBA variants across different patch sizes. Specifically, vanilla SSAMBA lags significantly behind W2V-LB, whereas BS-SSAMBA successfully closes this gap through the advantages introduced by the band-split method. Figure 2 shows the confusion matrix for BS-SSAMBA, with each entry representing the percentage of predicted classes normalized by true labels. The model struggles to distinguish active vocalization from silence, likely due to background noises in the audio. In speaker diarization, M-CDS and CN are often misclassified as the female caregiver, mainly due to the classifier’s bias towards majority class and the challenge of differentiating male child-directed speech from female speech. For vocalization classification, key challenges include distinguishing child-directed from adult-directed speech and misclassification between cry and fuss for the focal child.

METHOD	DEPTH	EMB	ACC	MACRO-F1	KAPPA
BS-SSAMBA [†]	12	192	0.813	0.569	0.641
BS-SSAMBA [†]	12	384	0.816	0.588	0.640
BS-SSAMBA	24	768	0.825	0.609	0.663

Table 4: Ablation study on model sizes. DEPTH and EMB respectively denote the number of mamba layers and the dimension of the patch embedding. [†] denotes smaller models uses 3x learning rate for the final classification head during finetuning, as in [19]. The models from top to bottom respectively have 6M, 16M, and 102M parameters during finetuning.

METHOD	N (PERCENT)	ACC	MACRO-F1	KAPPA
BS-SSAMBA	64 (25%)	0.808	0.561	0.627
BS-SSAMBA	128 (50%)	0.822	0.605	0.658
BS-SSAMBA	192 (75%)	0.825	0.609	0.663

Table 5: Ablation study on number of masked timesteps (N). N(PERCENT) represents the number of time steps masked for each band, as an integer and as a percentage of the $S = 256$ time steps.

3.4. Ablation Studies

In this section, we present several key ablation studies regarding BS-SSAMBA to evaluate the effect of different components and various hyperparameters.

Impact of Framewise Feature Conditioning: Table 3 presents our ablation study on framewise feature conditioning. We test BS-SSAMBA with and without framewise feature conditioning during finetuning with other configurations fixed. The results reflect the effectiveness of incorporating our proposed conditioning methods into the BS-SSAMBA framework.

Impact of Model Size: Table 4 presents our ablation study on model size. In general, larger BS-SSAMBA models with more layers and larger embeddings yield better results. However, performance differences across model sizes are small, and even the smallest 6M-parameter model achieves competitive results.

Impact of Number of Masked Patches: Table 5 presents our ablation study on different N, the number of masked patches in each band, ranging from masking 25% to 75% of patches on the temporal axis. In general, we observe an increase on the downstream infant-centered audio tagging performance when we mask out a larger portion of patches with our optimal setting at 75%.

4. Conclusion

We propose Band-Split SSAMBA, which integrates band-specific projections and a band-agnostic Mamba encoder into the Self-Supervised Audio Mamba framework. These modifications incorporate human knowledge of the spectrogram into the model, reducing sequence length for seq-to-seq modeling and facilitating efficient spectrogram representation learning with limited in-domain data. We evaluate our approach on unlabeled and labeled audio recordings from infant-centered home audio recordings. Our experimental results advance audio SSL techniques with limited in-domain data. Such advances are central to achieving a fuller understanding of this key development process: infants’ everyday vocal interactions with family members.

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